

Literature-implied full answer for arXiv:1212.6243

Dedekind completeness makes Archimedeaness automatic

Status. literature_implied_answer (full answer to the local question). The answer is an immediate consequence of a standard implication in vector lattice theory. The contribution of this packet is to notice that the source's hypotheses already contain that implication, not to claim a new order-theoretic theorem.

Exact source question. Rutsky's Proposition 18 assumes that Y is an *Archimedean Dedekind complete lattice*. Immediately after its proof, the paper asks whether the Archimedean assumption is necessary for the conclusion:

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example with some closed X_+ and open Y_+ . It is, however, not clear whether the assumption that Y is Archimedean is necessary for the conclusion of Proposition 18.

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The same paper defines Dedekind completeness to mean that every upper-bounded subset has a supremum.

Theorem. *Every Dedekind complete real vector lattice Y is Archimedean. Thus the answer to the source question is no: Archimedeaness is redundant under the remaining hypotheses of Proposition 18.*

Proof. Use exactly the source's definition of Archimedeaness. Suppose $x \in Y_+$ and $y \in Y$ satisfy

$$ny \leq x \quad (n = 1, 2, \dots).$$

The positive-part map of a vector lattice is order preserving, and it is positively homogeneous. Hence

$$ny^+ = (ny)^+ \leq x^+ = x \quad (n \geq 1).$$

The set $A = \{ny^+ : n \geq 1\}$ is therefore bounded above, so Dedekind completeness gives $s = \sup A$. Since $y^+ \geq 0$, this sequence is increasing, and its tail has the same supremum:

$$s = \sup_{n \geq 1} (n+1)y^+.$$

Translation by y^+ is an order isomorphism, so it preserves every existing supremum. Consequently

$$s = \sup_{n \geq 1} (y^+ + ny^+) = y^+ + \sup_{n \geq 1} ny^+ = y^+ + s.$$

Cancellation yields $y^+ = 0$, whence $y \leq 0$. This is precisely the Archimedean condition stated immediately before Proposition 18. □

Corollary. *In Proposition 18 of arXiv:1212.6243, delete the word “Archimedean” from “Archimedean Dedekind complete lattice.” The statement and proof remain valid. Likewise, in case (2) of Theorem 24, the separate condition that Y be Archimedean may be deleted because the theorem already assumes globally that Y is a Dedekind complete lattice.*

Proof. Apply the theorem to Y , then invoke the original results unchanged. □

Why this is literature-implied. The implication proved above is standard. For example, modern vector-lattice references routinely state that every Dedekind complete vector lattice is Archimedean. No additional Riesz–Kantorovich argument is needed.

Relation to the headline problem. The introduction also recalls the long-standing question whether an existing operator supremum must equal its Riesz–Kantorovich transform. That is a different question, and this run already has a packet recording Elliott’s 2019 negative answer under target arXiv:1501.01685. The present packet avoids double-counting that result and promotes only the distinct local question following Proposition 18.

Verification and search. The proof is symbolic and its five order-theoretic steps are audited in `verification.md`. Cheap run indexes were searched for the arXiv id, title, “Riesz–Kantorovich”, “Dedekind complete”, and “Archimedean.” A bounded exact-phrase/title search found the source and standard statements of the implication, but no later paper devoted to this local question.

Scope. This settles the necessity question only while retaining Dedekind completeness and the lattice hypothesis. It does not remove Dedekind completeness, answer the general operator-supremum problem anew, or address the source’s speculative quasi-interior-point extension.

References

- [1] D. V. Rutsky, *Remarks on the Riesz–Kantorovich formula*, arXiv:1212.6243 (2012; manuscript dated 2014).
- [2] M. Elliott, *The Riesz–Kantorovich formulae*, Positivity **23** (2019), 1245–1259.